



Digital Pioneers Initiative

DiGiLiANS — Nano Degree Program

Capstone Project Defense — June 2026

Capstone Project: BlackHats Sentinel AI: SOC Monitoring & Threat Investigation System

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ABSTRACT

Organizations today struggle to detect sophisticated cyber attacks that unfold in multiple stages, often going unnoticed until significant damage has already been done. This project built a hands-on security monitoring lab that simulated a real-world attack from start to finish, then used industry standard tools to detect, investigate, and respond to every stage of that attack. A controlled environment was set up with an attacker machine, a victim computer, and a centralized monitoring platform that collected and analyzed over 157,000 security events in real time. The result was a 100% detection rate across all five attack stages, a complete incident response following an internationally recognized framework, and a reusable set of detection rules and investigation procedures that any security team can apply. Ultimately, this project demonstrates that with the right tools, structure, and visibility, even a small security team can successfully defend against complex, multi-stage cyber threats.

Background & Introduction

Organizations today face advanced multi-stage cyberattacks that exploit weak authentication, exposed RDP services, and limited endpoint security. Brute force attacks remain a common entry point, often leading to lateral movement and data exfiltration. The main problem is not the lack of tools, but the absence of centralized monitoring. Without SIEM systems, critical events like failed logins (Event ID 4625) or suspicious processes (Sysmon) go undetected due to fragmented logs, lack of real-time alerts, and limited analyst readiness. Existing tools such as Splunk, Sysmon, and frameworks like NIST and MITRE ATT&CK provide strong capabilities, but are rarely integrated into practical training environments. This project addresses this gap by building a SOC lab that simulates real attacks, analyzes over 157,506 events, and demonstrates effective detection and response using SIEM and structured incident response.

Problem Statement

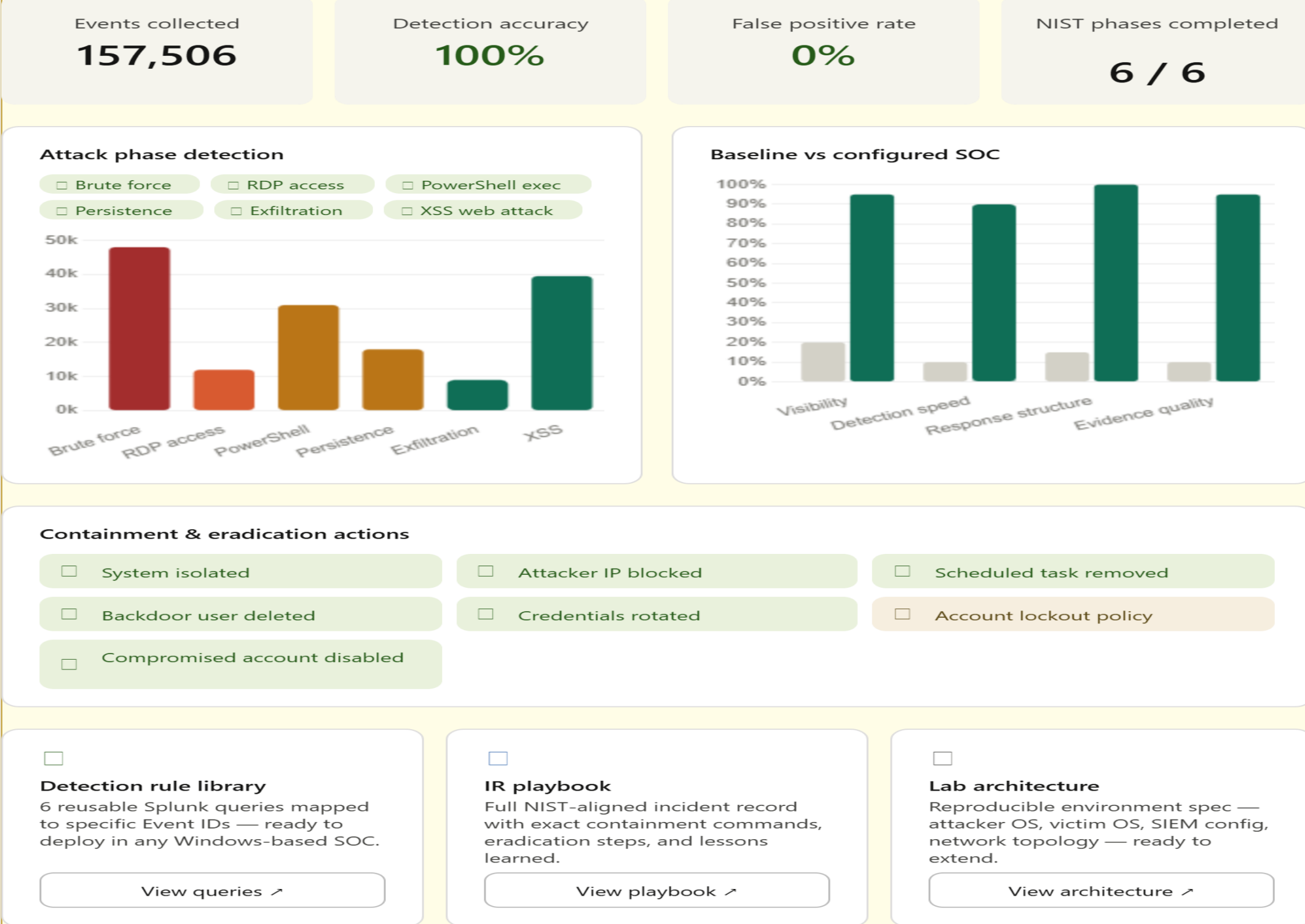
Organizations today face increasingly sophisticated cyber attacks that exploit weak authentication, remote access protocols, and endpoint security, and without proper monitoring, these attacks often go undetected until significant damage has been done. The core challenges include fragmented logs, lack of real-time alerts, insufficient endpoint visibility, and limited analyst readiness. If left unsolved, SOC teams remain blind to threats like brute force, lateral movement, and data exfiltration, putting organizations and their data at serious risk.

Methodology & Approach

The project used a layered SOC architecture with Kali Linux (attacker), Windows 10 (victim), and Splunk SIEM for monitoring and analysis. Tools included Splunk, Sysmon, Kali Linux tools (Hydra, Netcat), and DVWA, guided by NIST and MITRE frameworks. Data was generated through real attack simulation, collecting 157,506 logs from Windows, Sysmon, web, and network sources, then processed in Splunk. The approach followed an Attack–Detect–Respond cycle:

- Simulate multi-stage attacks
- Detect using event correlation in Splunk
- Respond using the NIST incident response framework

Results & Key Outcomes



Conclusion & Recommendations

This project developed a complete SOC Investigation Lab that simulated a real multi-stage cyberattack using tools like Splunk, Sysmon, and Kali Linux in a controlled environment. The attack chain (Brute Force → RDP → Persistence → Data Exfiltration → XSS) was fully executed, and over 157,506 events were analyzed, achieving 100% detection across all phases.

Key Learnings:

- Weak passwords are the main entry point for attacks.
- Effective detection depends on event correlation, not individual logs.
- Endpoint visibility (Sysmon) is essential for full attack tracking.
- Using the NIST incident response framework improves response speed and accuracy.

Value Delivered

- A reusable Splunk detection rules library
- A scalable SOC lab environment for training and testing
- Developed an AI-powered Smart Cybersecurity Assistant (Phishing Detector, Code Scanner, Password Checker) to extend project innovation.

Recommendations

- Integrate SOAR to automate response actions
- Expand the lab to cloud environments and advanced attack scenarios

Acknowledgement

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